

Locally Editing Steady Fluid Flow via Controlling Repulsive Forces from Terrain

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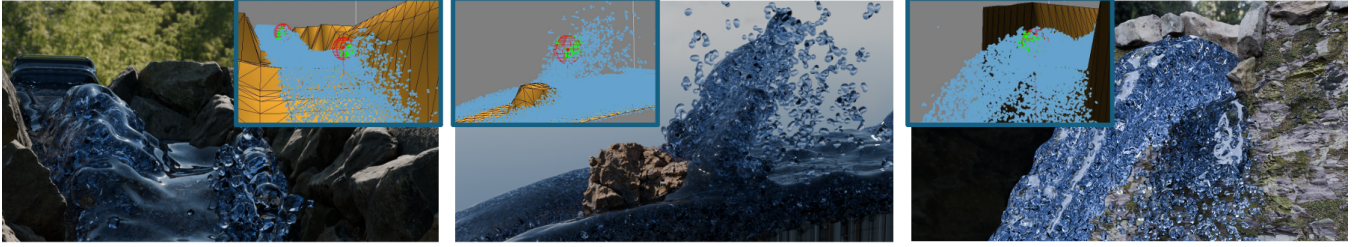


Figure 1: Three examples of water flows (River, Rock, Waterfall) edited by our method. The insets are preview images with our interactive editing tool. The red spheres indicate the target regions. Water successfully passes through the target regions.

Abstract

This paper presents a novel control method for steady fluid flows, such as rivers and waterfalls, simulated using SPH. Our system allows the user to place target points through which the fluids should pass, and the fluid flows can be controlled with these points. This is achieved through the control of repulsive forces on terrain surfaces. We detect positions on the terrain that affect fluid particles approaching the target points, and repulsive forces at these positions are automatically adjusted through a feedback control mechanism. Our method achieves the local control of steady fluid flows, and we demonstrate its effectiveness through several examples.

Keywords

fluid simulation, SPH, fluid control, feedback control

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1 Introduction

Physically-based simulations have been widely used for creating realistic animations in entertainment applications, such as movies and video games. Among them, fluids are one of the important elements that enhance the realism and immersion by visual effects

in synthetic scenes. In this paper, we focus on the simulation of steady fluid flows like rivers and waterfalls. One of the difficulties with fluid simulation is predicting and controlling its motion due to its nonlinearity, which makes it challenging for animators and artists to create the desired visual effects. To address this problem, extensive research has been conducted on fluid control [Pan et al. 2013; Schoentgen et al. 2020]. However, most previous control methods are designed for unsteady fluid flows and are not suitable for controlling steady flows, such as rivers and waterfalls.

We propose a novel control method for steady fluid flows. Our system controls fluid flows, simulated based on SPH (Smoothed Particle Hydrodynamics) [Bridson 2008], through repulsive forces generated when fluids collide with terrain. Users place target points, and then repulsive forces are controlled such that fluids pass through these target points. This control is achieved by repeating two processes: detecting positions to update repulsive forces and automatically adjusting the magnitudes of repulsive forces at the detected positions. To adjust the repulsive forces, we utilize the feedback control mechanism. Using our system, steady fluid flows can be locally edited through a simple operation.

2 Our Control Method

An overview of our control method is shown in Fig. 2. The basic idea is to apply artificial external forces (control forces) that correspond to the repulsive forces caused between fluids and terrain, so that fluids pass through target regions defined by spheres centered at the target points. The terrain is sampled at discrete locations, and the control forces $\mathbf{f}_{i,k}^{ctrl}$ are applied at each sample point $\mathbf{q}_k$ ($k = 0, \dots, N_t - 1$), where N_t is the number of sample points. The control forces are updated at each frame of the simulation. First, we search for SPH particles near the target points and detect sample points (called update points) on the terrain that affect the motions of these

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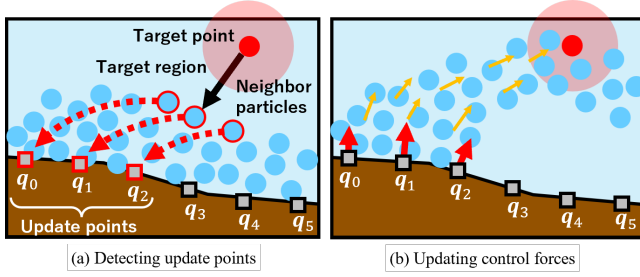


Figure 2: Overview of our control system.

particles (Fig. 2(a)). Then, control forces at these update points are updated through feedback control (Fig. 2(b)). The details of each process are described below.

2.1 Calculating repulsive forces

Let us denote the position of a SPH particle by $\mathbf{p}_i (i = 1, \dots, N_p)$ flowing on the terrain, where N_p is the number of the particles. When particles collide with the terrain, repulsive forces work in the normal direction to the terrain, creating a natural flow. In our method, the repulsive force $\mathbf{f}_{i,k}^{rep}$ that particle $\mathbf{p}_i$ receives from sample point $\mathbf{q}_k$ is defined as follows:

$$\mathbf{f}_{i,k}^{rep} = (d_{i,k}\xi - \eta(\mathbf{n}_k \cdot \mathbf{v}_i))\mathbf{n}_k + \mathbf{f}_{i,k}^{ctrl}, \quad (1)$$

where $d_{i,k} = \|\mathbf{p}_i - \mathbf{q}_k\|$, ξ and η are the elasticity and damping coefficient for the collision, respectively. $\mathbf{n}_k$ is the normal vector at sample point $\mathbf{q}_k$ on the terrain, and $\mathbf{v}_i$ is the velocity of particles $\mathbf{p}_i$. Our method automatically adjusts the magnitude of the repulsive force, $\mathbf{f}_{i,k}^{ctrl}$, to control the flow.

2.2 Specifying target points

The user can place multiple target points (the red points in Fig. 2), and a sphere-shaped region (the red shaded areas in Fig. 2) is defined around the target point. The radius of each target region is specified by the user. Our system controls the flow such that SPH particles near each target point pass through these target regions.

2.3 Detecting update points

When a particle collides with the terrain, it receives a repulsive force. The particle then collides with its surrounding particles, causing the repulsive force to propagate among them. We use this idea to detect the update points. That is, we detect the terrain sample points where the repulsive forces propagated to the particles near the target point are the strongest. Such points are shown as $\mathbf{q}_0$, $\mathbf{q}_1$, and $\mathbf{q}_2$ in Fig. 2(a).

To implement this idea, each particle records a list of propagated repulsive forces received from each terrain sample point. That is, for particle $\mathbf{p}_i$, we record list $F_{i,k}$ representing the magnitude of the repulsive force propagated from terrain sample point $\mathbf{q}_k$. We refer to $F_{i,k}$ as the repulsive force list and use it to identify the update point that has the most significant effect on each particle.

$F_{i,k}$ is updated when collisions with the terrain or other particles occur. When particle $\mathbf{p}_i$ collides with a sample point $\mathbf{q}_k$, $|\mathbf{f}_{i,k}^{rep}|$ is substituted for $F_{i,k}$. When particle $\mathbf{p}_i$ collides with its neighboring particles, the force recorded in the repulsive force list $F_{i,k}$ is distributed evenly among all the neighbor particles $\mathbf{p}_j$: $F_{j,k} \leftarrow F_{j,k} + \frac{F_{i,k}}{N}$, where N is the number of the neighboring particles. After distributed, all elements of $F_{i,k}$ for particle $\mathbf{p}_i$ are initialized to zero.

2.4 Adjusting control forces

Control force $\mathbf{f}_{i,k}^{ctrl}$ at sample point $\mathbf{q}_k$ for particle $\mathbf{p}_i$ is calculated as follows:

$$\mathbf{f}_{i,k}^{ctrl} = J_k \exp(-\zeta d_{i,k})\mathbf{n}_k, \quad (2)$$

where J_k is a control coefficient, and ζ determines the range of influence of the control force. J_k is automatically adjusted as a control variable through the feedback control mechanism. We utilize only the P (proportional) controller, which updates the control variable in proportion to the objective function. Our objective function for J_k is calculated with the corresponding target point and its neighboring particles:

$$J_k = \frac{K_p}{N_k} \sum_{n_k} \text{sgn}(d_{tar,k} - d_{n_k,k}) \|\mathbf{p}_{tar} - \mathbf{p}_{n_k}\|, \quad (3)$$

where $\text{sgn}(d)$ is the sign function, $\mathbf{p}_{tar}$ is the position of the target point, and K_p is the proportional gain. $\mathbf{p}_{n_k}$ represents the particles near the target point corresponding to the update point $\mathbf{q}_k$, and N_k is the number of such particles. J_k is updated at regular frame intervals, as a certain amount of time is required for the repulsive force to propagate through the particles near the target points.

3 Results

Fig. 1 shows three examples of water flow edited by our method. The preview images are shown in the inset. The water successfully passes through the target regions, and realistic animations with the desired flows are created (see our supplemental video). Our method is very efficient and the optimization process converges quickly. This allows the user to interactively design the water flow.

4 Conclusions

We proposed a method to locally edit steady fluid flows. The magnitude of the control forces defined at terrain points was sequentially updated so that the fluid would pass through the regions near the target points specified by the user. This allowed us to locally control the behavior of the steady state flows like rivers and waterfalls. Future work includes developing an optimization method that also considers the direction of the control forces.

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